

Kabir Singh

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ACADEMIC EDUCATION

- Albert Dorman Honors College (ADHC) & Jordan Hu College of Sciences & Liberal Arts** (JHCSLA), New Jersey Institute of Technology, Newark, New Jersey, USA Sep '23 – May '27 (Expected)
- Bachelor of Science; Major – **Mathematical Biology**
 - Minors – **Biomedical Engineering in Technology, Philosophy, History**
 - ADHC Honors Tracks – **Medical Humanities, Global Studies**
 - *Academic Excellence, Honors Merit, Honors Residential Scholarship Recipient (Tuition + Board)*
- Faculty of Engineering, Lunds Tekniska Högskola, **Lunds Universitet**, Lund, Sweden Jan '25 – Jun '25
- Fellow, Institute of International Education's Global Engineering Education Exchange (Global E3)

RESEARCH

◇ *Computational and Mathematical Biology* ◇

- Undergraduate Researcher, **Computational Biology Team Lead** at KumarLab. PI: Dr. Vivek Kumar, University of Houston (Prev. NJIT) Oct '23 – Present
- Lead undergraduate and graduate students on designing and developing peptide-based therapeutics
 - Started at NJIT and continued computational work with lab relocation to Houston
 - Taught 15+ associates in lab
 - Received > \$15,000 in direct funding for projects
- Undergraduate Researcher, Computational Bionanotechnology. PI: Dr. Stefano Angioletti-Uberti, Imperial College London Jun '26 – Present
- **BAGEL: Biomolecular Algorithm for Guidance in Energy Landscape**
 - Computational modelling of soft-matter and biomaterial systems
 - Applying statistical mechanics, molecular dynamics, and machine-learning methods to problems in enzyme design and broader protein design
- Undergraduate Researcher, Mathematical Neuroscience. PI: Dr. Horacio Rotstein, NJIT Sep '26 – Present
- Deriving reduced mathematical descriptions of biophysical neuronal models to characterize their dynamic properties
- Undergraduate Researcher, **Mathematical Modeling**. PI: Dr. James Maclaurin, NJIT Oct '25 – May '26
- Applying Hawkes self-exciting point process to model secondary nucleation dynamics in amyloid formation

◇ *Intellectual and Cultural History* ◇

- Emerging Scholars Research Fellow**, Early Modern Sikh History. Mentor: Satnam Singh, Harvard Sikh Center Sep '26 – Present
- *Dissemination and the making of Khalsa authority in a contested panth*: examining how the Khalsa asserted dominance over the wider Sikh world within decades of its founding, against rival claimants including the Bandai Khalsa, Minas, and Ram Raiyas
 - Arguing that the consolidation of “authentic Sikh teachings” through katha was instrumental to that dominance; working from Gurmukhi and Persian primary sources, including works attributed to Bhai Mani Singh, hukamname, and rahit-nama literature
- Undergraduate Researcher, **History of Medicine & Bioethics**. PI: Dr. Stephen Pemberton, NJIT Sep '26 – Present
- Synthesizing legal, clinical, and cultural developments in American end-of-life care from the late twentieth century to the present
 - Developing original scholarship for submission to a peer-reviewed humanities journal

Undergraduate Researcher, **Digital History & Urban Studies**. PI: Dr. Louis Hamilton, NJIT Jan '24 – May '24,
Jan '26 – May '26

- Digital mapping of how religious institutions shaped urban form in medieval and early modern Italy; field research conducted in Rome and Florence
- *Power vs. the People* (HIST 328; 18 sites): layered Counter-Reformation Marian shrines against Julius II's 1508 papal axis on the Via Giulia
- *Processions and Power* (HIST 329; 22 sites): traced how Florence's expansion turned liturgical processions into contested displays of ecclesiastical and household prestige

◇ *Emergency Care and Health Services* ◇

Undergraduate Researcher, Collegiate Emergency Medical Services. Dr. Chang Yaramothu, NJIT Sep '25 – Present

- Studied the resources of ~50 US collegiate EMS squads, examining their organizational setups and capabilities in relation to their effectiveness

FUNDING

Undergraduate Research Opportunities Programme (UROP), Imperial College London Jun '26 – Jul '26

- Competitive research bursary awarded through the Department of Materials, Faculty of Engineering
- Project: Using protein language models to mimic natural ligands. Supervisor: Prof. Stefano Angioletti-Uberti
- Personal compensation included

NSF I-Corps Regional Program, Northeast Hub May '26 – Jun '26

- Co-Founder & Technical Lead, Axioma Therapeutics
- Selected into competitive NSF cohort; led customer discovery and lean-startup validation for cyclic peptide therapeutics targeting degenerative eye diseases
- Project funding awarded to support customer-discovery activities

Grace Hopper AI Research Institute, Computational Protein Design May '25 – Aug '25

- NJIT grant to develop a course on computational protein design
- Teaching Assistant

The Honors College Moonshot Grants for Engaged Scholarship Jun '25 – Jul '25

- Endowed award to develop interdisciplinary projects that address important social and cultural questions
- Designing Male Contraceptive
- Award providing labs with funds to support undergraduate-led projects
- Personal compensation included. One recipient per annum

Male Contraceptive Initiative (MCI) Jan '25 – Jun '25

- Award by the Youth Advisory Board providing funding to support grantee-led project
- Designing Male Contraceptive
- Personal compensation included

NJIT Undergraduate Research & Innovation (URI) Oct '24 – May '25

- Student Seed Grant award for a Phase-2 longitudinal project on application-based research funding lab to support undergraduate-led project
- Developing cyclic peptide therapeutic for Multiple Sclerosis

Travel Award – National Contraception Meeting May '25

- Awarded by the Male Contraceptive Initiative for travel to Durham, NC, supporting a conference presentation

NJIT Honors Summer Research Institute (HSRI) Jun '24 – Jul '24

- Grant award by ADHC and URI providing funds to support undergraduate-led projects
- A combination of Multiple Sclerosis project and Male Contraceptive project
- Personal compensation included

PUBLICATIONS

Peer-Reviewed

- C. Heffernan, O. West, **K. Singh**, J. Dodd-o, M. James, A. Baradi, K. Rybak, A. Lateef, S. Elkabes, V. Kumar. A novel cyclic peptide that blocks human autoimmune antibodies for the treatment of multiple sclerosis. *European Journal of Medicinal Chemistry*, 119173 (2026). doi:10.1016/j.ejmech.2026.119173

Manuscripts in Preparation

- **K. Singh**, C. Heffernan, O. West, J. Dodd-o, V. Kumar. Machine learning for rational design of a ***** targeting cyclic peptide-based male contraceptive. Publication [in preparation], **Patent application** [in preparation], 2026
- J. Lau, D. Yun, **K. Singh**, A. Patananan, N. Jadavji. Academic Publishing as a Launchpad: How Publishing Research in an Undergraduate Journal Shapes Future Careers. Publication [in preparation], 2026

PRESENTATIONS

M. Kotta, G. Victorio, G. Karmacharya, **K. Singh**, N. Giovannetti, B. Shadpoor, J. Dodd-O, V. Kumar. Putative computational design of novel inhibitors to modulate de-novo targets for CB1 receptor.

UH (Undergraduate Research Day)

Poster + Podium 2026

K. Singh, C. Heffernan, J. Dodd-O, V. Kumar. Development of a novel male contraceptive using cyclic peptides to inhibit fertilization.

AAP/ASCI/APSA Joint Meeting

Poster

2026

NJIT AI Exploration Day

Poster

2026

NJIT ADHC (Honors Interdisciplinary Research Forum) **First Place**

Poster

Fall 2025

NCM (National Contraception Meeting)

Poster

2025

NJIT ADHC (Honors Interdisciplinary Research Forum)

Poster

Spring 2025

NJIT (Dana Knox Research Showcase)

Poster

2025

NJIT (Undergraduate Research & Innovation)

Abstract + Podium

2025

BMES (Biomedical Engineering Society) [Co-Presenter]

Abstract + Poster

2024

NJIT (Undergraduate Research & Innovation)

Abstract + Podium

2024

K. Singh, J. Lau, H. Gerber, A. Alibilli, L. Goralsky. Mitigating Knowledge Drain: A Case Study in Optimizing Workflow and Training in a High-Turnover, International Peer-Reviewed Undergraduate Science Academic Journal.

NCUR

Poster 2026

ACADEMIC SERVICE

◇ Editorial and Peer Review ◇

22nd Editor-in-Chief, Associate Editor (Former), Journal of Young Investigators

Dec '24 – Present

- Manage timely publication of high-quality scientific content
- Develop initiatives to further the journal's mission
- Former specialization in Biology, Physical Sciences, and Social Science Departments

Grant Reviewer, MCI YAB Undergraduate Project Grant

Dec '25 – Jan '26

◇ Teaching and Mentorship ◇

Research Lead, Governor's STEM Scholars, Research & Development Council of New Jersey

Sep '26 – May '27
(Expected)

- Guiding a team of high school Scholars through a year-long research and engineering design project anchored to a UN Sustainable Development Goal and addressing a New Jersey community need
- Overseeing team meetings, deliverables, timeline, and budget, and preparing the team's final presentation across four statewide conferences on STEM in government, academia, and industry
- Appointment within a public-private partnership of the Office of the Governor, NJ Department of Education, and NJ Office of the Secretary of Higher Education; \$1,000 completion stipend

Teaching Assistant, Honors First Year Seminar

Sep '24 – Dec '24,
Sep '25 – Dec '25

Teaching Assistant, Grace Hopper AI Research Institute

May '25 – Aug '25

- Computational Protein Design

◇ National and Professional Service ◇

Research Chair , American Association of Physicians of Indian Origin (AAPI) PreMed Board	Jun '25 – Present
– Lead committee and organize events to achieve research-related objectives on a national level to support students, researchers, physicians, and more of an Indian origin	
Board Member , MCI Youth Advisory Board (YAB)	Aug '25 – Present
Member, Pre-Medical Ad Hoc Committee, APSA	Mar '26 – Present
Member, Goldwater Ambassadors Program	Aug '25 – Present
– Selected into competitive process as a mentee, mentored by Goldwater Scholars	
– Program specially designed for students from underrepresented or underserved backgrounds	
Member, ARTISAN Program	Apr '25 – Present
– Inducted in the NSF ART Integrated Student Ambassador Network (ARTISAN) program, based off commitment to translational research in my work	
◊ <i>University Service</i> ◊	
Member, Student Conduct Board	Dec '24 – Present
– Serve as adjudicator on academic and behavioral cases in student conduct hearings, along with other members of the board, including faculty and NJIT administration	
Chair, Chair-in-Training (Former), Highlander Integrity Council	Sep '24 – Present
Member, Inclusive Excellence Student Advisory Board	Feb '25 – Present
Member, Multifaith Council	Jan '25 – Present
Organizer, 'It's Different for Girls' Documentary Screening & Panel, NJIT ADHC & CSLA Co-Sponsored Colloquium	Mar '26
NJIT Representative, BioNJ 33 rd Annual Dinner Meeting & Innovation Celebration	Feb '26
Steering Committee Lead, ADHC First Year Retreat	Mar '25 – Aug '25
Advisor, President (Former), Secretary (Former), Honors Hall Council	Feb '24 – May '25
ADHC Representative, National Collegiate Honors Council Annual Conference	Nov '24

LECTURES & FEATURES

- Invited Speaker (forthcoming). Student roundtable with Margo Chaly, New Jersey's Secretary of Education. NJIT, 2026
- Invited Speaker. Presentation on interdisciplinary research in medicine and healthcare to NJIT leadership, including the Interim President and Interim Provost. University Advancement Dinner, NJIT, 2026
- Lecture Series. Ethics in Science and Research. Science Mentorship Institute, 2026
- Session Moderator. Student Oral Presentation (Communications). NCUR, 2026
- Featured Presenter. 'Navigating peer review and the student journal landscape'. "Shaping Asia's Future: Development, Innovation, and Society in Transition" organized by Nazarbayev University, Asian Universities Alliance, 2026
- Featured Panelist. 'It's Different for Girls' Documentary Screening & Panel. NJIT ADHC & CSLA Co-Sponsored Colloquium, NJIT 2026
- NJIT Makes: Undergraduate Research & Innovation, NJIT Promotional Material. NJIT, 2026
- Invited Speaker. Presentations to the NJIT Board of Trustees and the President's Town Hall with squad leadership, at the request of NJIT President Teik Lim. NJIT First Aid Squad, 2025
- Albert Dorman Honors College, Honors in Action 2025 (Research, Award, and Study Abroad Experience). ADHC Publication. NJIT, 2025
- Nayib Morán. '150+ Honors Scholars Showcase Findings at Interdisciplinary Research Forum'. Article Feature. NJIT News, 2025
- Featured Speaker. ADHC Advising Day. NJIT, 2024 & 2025
- Albert Dorman Honors College, Honors in Action 2024 (Research and Study Abroad Experience). ADHC Publication. NJIT, 2024
- Gurnoor Kaur. 'Undergraduate Student Researchers on Campus'. Article Feature. NJIT (Student Newspaper – The Vector), 2024
- K. Singh. 'Introduction to PyMOL and Linear Peptide Design'. Lecture. NJIT (Genetically Engineered Machines), 2024

CERTIFICATIONS & LICENSES

- New Jersey EMT License
- ACS "Stop the Bleed" Certification & Instructor License
- OSHA General Industry Safety & Health Certification

- CITI Biomedical Research Certification
- CITI RCR Basic Certification
- American Heart Association BLS & CPR for Adult/Child/Infant Certification

LEADERSHIP

Ethics Coordinator, Mentor (Neuro + Bio), Ethics Curriculum Dev (Former), Admission Officer, Symposium Committee Organizer, Grant Writer, Science Mentorship Institute (Sci-Mi)	May '24 – Present
Public Relations Chair, EMT, NJIT First Aid Squad	Oct '25 – Present, Jan '24 – Present
Khalsa School Virsa (Sikh History) Teacher, Sikh Sabha of New Jersey	Sep '20 – Present
Youth Camp Organizer & Coordinator, Sikh Sabha of New Jersey	Aug '21 – Present
Content Manager, The Mardana Project	May '26 – Present
Stop the Bleed Instructor, NJIT	Oct '25 – Present
Stop the Bleed Instructor (Hybrid English–Punjabi), NJ Gurdwaras, NJ Office of Homeland Security & Preparedness (NJOHSP) & Sikh Youth Alliance	Jan '26 – May '26
Youth Camp Counselor, Guru Nanak Sikh Heritage	Aug '21 – Aug '25
Vice President, President (Former), Events Coordinator (Former), Sikh Association of NJIT (SANJIT)	Dec '23 – Present
President, Treasurer (Former), Co-Founder, NJIT MedicZero	Nov '23 – May '26
Co-Founder, Delta Epsilon Mu (Pre-medical Professional Fraternity)	Feb '24 – May '25
Fellow, Honors Summer Research Institute Fellowship	Jun '24 – Jul '24, Jun '25 – Jul '25
EMT Senior Squad Member, Cadet Crew Chief (Former), Marlboro First Aid Squad	Apr '22 – Jan '25
Fellow, No Patient Left Behind Fellowship	Jun '24 – Aug '24
Young Scholar, American College of Cardiology	May '23 – Apr '24

References and letters of recommendation available upon request.